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https://doi.org/10.1007/s12633-024-03117-2

RESEARCH

Comparative Analysis of Cure Behaviors, Mechanical Properties, and Swelling Resistance in EPDM/SBR Composites with HNTs, APTES-Modified HNTs, and RH-Modified HNTs

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Received: 17 May 2024 / Accepted: 8 August 2024

Abstract
Rubber blending is a prominent technique for enhancing properties in final rubber products. This study investigates the interplay of filler concentration and surface modification in EPDM/SBR blend composites with halloysite nanotubes (HNTs). The effects of γ-Aminopropyltriethoxysilane (APTES) and resorcinol-hexamethylenetetramine (RH) modifiers were examined. Comparing rubber blend composites with modified and unmodified HNTs, the findings reveal significant enhancements using RH-modified HNTs. These composites outperform those with APTES-modified and unmodified HNTs, notably improving mechanical properties. The addition of fillers increases crosslink density and filler-rubber interaction, reducing mole percent uptake. These trends result in significantly improved abrasion resistance in the composites. FESEM images show that RH-modified HNTs have superior distribution compared to APTES-modified and unmodified HNTs, highlighting their effective interaction and dispersion. These findings can guide the optimization and production of outdoor applications.

Keywords EPDM/SBR · HNTs · APTES-modified HNTs · RH-modified HNTs · Mechanical Properties · Swelling Resistance

1 Introduction

Polymer nanocomposites represent an emerging category of advanced materials that has experienced rapid evolution over the past two decades [1–5]. The heightened attention directed towards polymer nanocomposites stems from their distinctive attributes, primarily rooted in the considerable surface area of the nanofillers. The extensive surface area leads to highly beneficial interactions with the polymer matrix, leading to enhancements in properties surpassing those of the pure polymer matrix and outperforming conventional composites even with significantly lower filler loadings [6]. This unique advantage has propelled the exploration of polymer nanocomposites as a cutting-edge avenue for achieving exceptional material performance with relatively minimal filler incorporation.

Extensive literature underscores the pivotal significance of nanoparticle dispersion and their interfacial characteristics within the surrounding matrix as fundamental factors for achieving desired material attributes. Numerous elements have been scrutinized within the realm of polymer nanocomposites, including carbon nanotubes (CNTs), halloysite nanotubes (HNTs), clay platelets (nanoclay or organoclay), and graphene oxide (GO) particles [7–9]. More specifically, the search for precise control over the exfoliation and intercalation processes has emerged as a critical emphasis in nanocomposites including graphene and clay particles. Yet, the endeavor towards achieving such precise manipulation invariably drives up fabrication costs and introduces heightened uncertainties into performance outcomes [10–16]. Furthermore, the potential

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Published online: 14 August 2024